JAKE CAPMEIL.

STUDENT RESEARCHER · ROBOTICS & SOFTWARE

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01 EDUCATION

Hunter College High School · New York, NY — Expected June 2028

- **Coursework:** AP CS A, Honors CS, Extended-Honors Math (Precalculus, Statistics, Probability), Honors Physics, AP French
- **Science Research Seminar:** selective seminar entered by qualifying exam; machine learning, advanced Python, and statistics
- **Awards:** Two-time Gold Medalist, National French Grand Concours (native-speaker division)

02 RESEARCH EXPERIENCE

Research Intern · Control/Robotics Research Laboratory (CRRL), NYU Tandon — Aug 2026 – Present

- Built **CupFlow** under Prof. Farshad Khorrami: a robot-learning data pipeline on a Franka Research 3 arm that records a full multimodal episode for every pick-and-place; **900+ episodes** collected across 150 runs
- Synchronized ZED stereo RGB-D, a wrist RealSense, 50 Hz proprioception, and Vicon mocap on one clock, with hand-eye calibration
- Wrote teleoperated, scripted, and fully autonomous collection modes, plus the export to fixed-rate RLDS datasets in LIBERO's schema for vision-language-action policy training

CS Scholars Research Mentee · Carnegie Mellon University — Summer 2026

- Selected for a fully funded, merit-based CMU School of Computer Science program (~4% admission rate)
- Authored *Threshold Tuning Under Class Imbalance for Predicting Student Preference in AI-Assisted Tutoring* and presented it at the closing symposium
- Benchmarked three classifiers with k-fold cross-validation, optimized for MCC, and tuned thresholds to correct severe class imbalance

03 PROJECTS

Atlas of Crowns · Historical Strategy Game for iOS & Web · atlasofcrowns.com — 2026 – Present

- Designed, built, and published solo: play any nation from 476 onward on a real-world map, powered by a deterministic simulation, a canvas map renderer, online multiplayer, and an iOS build
- Marketed it end to end with a short-form video ad campaign produced in code (Remotion) and a creator beta-access program

Bloom · Full-Stack Habit-Tracking Web App — 2026 – Present

- Shipped on Cloudflare Workers with Google sign-in, Supabase, an AI coach, and a tree that grows with streak consistency

Khromate · Generative Art Platform · khromate.com — 2026 – Present

- Built the full stack, from generation engine to public gallery, for royalty-free art made from coded algorithms

Aspect Ratio Effects on Flight Performance in Free-Flight Gliders — 2024 – 2025

- Built a modular sailplane with swappable equal-area wings of varying aspect ratio; flight-tested lift, stability, and glide

Bathymetric Control of Breaking Wave Height in Nearshore Environments — 2023 – 2024

- Built a wave tank with interchangeable seafloors; measured how slope, depth, and contour change shoaling and wave height

04 LEADERSHIP & SERVICE

Board Member & Web Designer · Forward Progress USA · forwardprogressusa.com — 2026 – Present

- Created the nonprofit's unified design system and run its website front to back (Cloudflare Workers, sign-up backend); help produce content and volunteer at events

Vice President & Founding Member · Interact Club of Chinatown (Rotary) — May 2025 – Present

- Co-founded the youth arm of Chinatown's Rotary Club; raised $1,500+ in year one, grew to 40+ members; run its events and site

Varsity Captain · Cross Country & Track — 2025 – Present

- Placed 2nd in the PSAL Sophomore 4×400m and 4th in the 800m in a first varsity season; competed at Cities
- Lead workouts, mentor underclassmen, and manage race-day logistics while training seven days a week

Volunteer Surf Instructor · Laru Beya Collective — Summers 2024 – Present

- Teach surfing and water safety to underserved NYC youth through a nonprofit surf program

English Peer Tutor & Parish Volunteer — 2024 – Present

- Tutor peers in grammar, structure, and analysis (chosen among the grade's top writers); run biweekly refugee clothing drives

05 SKILLS

Programming & ML	Python, TypeScript, pandas, NumPy, supervised learning, neural networks, data pipelines
Robotics	ROS 2, Franka Research 3 / FR3Py, Vicon motion capture, ZED / RealSense, hand-eye calibration, RLDS, Docker
Full-Stack	Cloudflare Workers, Supabase, Capacitor (iOS), HTML/CSS, GitHub, Render, Netlify
AI Engineering	Claude Code, agent workflows, CLAUDE.md specs, in-app AI features, end-to-end deployment
Lab & Other	PCR, gel electrophoresis, micropipetting · Logic Pro, Affinity · **Languages:** English, French (fluent)